

Problem 1) $T = \frac{1}{2} m \dot{x}^2 = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t - \delta) = \frac{1}{2} k A^2 \cos^2(\omega t - \delta)$

The potential energy gain is the same as the amount of work done by the force:

$$-dW = -F dx = k dx = dU$$

$$\Rightarrow U = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \sin^2(\omega t - \delta)$$

$$\Rightarrow E = T + U = \frac{1}{2} k A^2 [\cos^2(\omega t - \delta) + \sin^2(\omega t - \delta)]$$

$$\Rightarrow E = \frac{1}{2} m A^2 \omega^2$$

Problem 2) The tangential component of $-mg$ is,

$$-mg \sin \theta = m(l\ddot{\theta})$$

This eqn. is not analytically solvable.

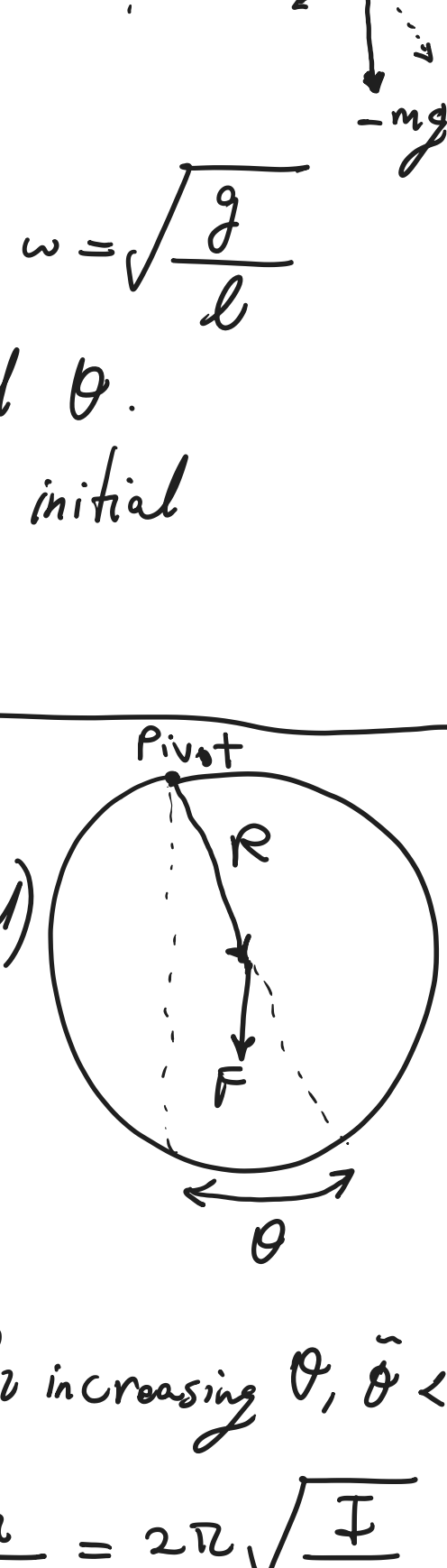
We must make a small angle approximation.

$$\lim_{\theta \rightarrow 0} \sin \theta = \theta$$

$$\Rightarrow \ddot{\theta} + \omega^2 \theta = 0; \quad \theta \rightarrow 0 \quad \text{where } \omega = \sqrt{\frac{g}{l}}$$

$$\Rightarrow \theta(t) \approx A \cos(\omega t + \phi) \quad \text{for small } \theta.$$

A and ϕ are unknown constants determined by the initial conditions of the problem.



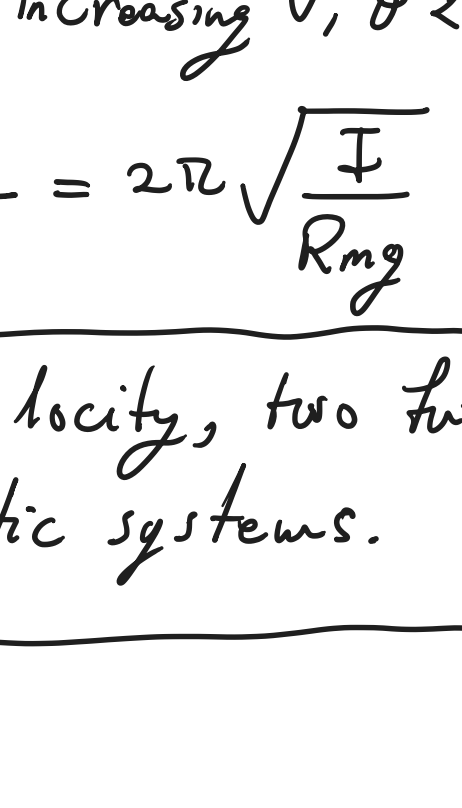
Problem 3) Let the rotational inertia of the sphere be I . (In reality $I = \frac{7}{5} m R^2$, but it is not needed)

The torque on the sphere is $N = R \times F$

$$|N| = R F \sin \theta = R m g \sin \theta \approx R m g \theta \quad \text{for } \theta \rightarrow 0$$

Thus, the eqn. of motion is, $I \ddot{\theta} = -R m g \theta$ (because for increasing θ , $\ddot{\theta} < 0$)

$$\Rightarrow I \ddot{\theta} + R m g \theta = 0 \Rightarrow \omega = \sqrt{\frac{R m g}{I}} \Rightarrow \tau = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{R m g}}$$



Problem 4) No. If so, for the same position and velocity, two future would be possible, which is impossible for deterministic systems.

Problem 5) For the simple harmonic motion:

$$x(t) = A \sin(\omega_0 t - \delta); \quad \dot{x}(t) = A \omega_0 \cos(\omega_0 t - \delta)$$

$$\Rightarrow \frac{x}{A} = \sin(\omega_0 t - \delta); \quad \frac{\dot{x}}{A \omega_0} = \cos(\omega_0 t - \delta)$$

$$\Rightarrow \left(\frac{x}{A}\right)^2 + \left(\frac{\dot{x}}{A \omega_0}\right)^2 = 1 \quad \text{This is the eqn of an ellipse whose}$$

$$\text{semi axes are: } A = \sqrt{\frac{2E}{k}} \quad \text{and} \quad A \omega_0 = \sqrt{\frac{2E}{m}}$$

Problem 6) A damped oscillator is described by $x(t) = e^{-\beta t} \cos(\omega_1 t - \delta)$

$$\Rightarrow \dot{x}(t) = -\frac{1}{2} A e^{-\beta t} \left[\beta \cos(\omega_1 t - \delta) + 2\omega_1 \sin(\omega_1 t - \delta) \right]; \quad \omega_0 = \sqrt{\frac{k}{m}}$$

$$\Rightarrow U = \frac{1}{2} k x^2 = \frac{1}{2} m A^2 \omega_0^2 e^{-\beta t} \cos^2(\omega_1 t - \delta)$$

$$T = \frac{1}{2} m \dot{x}^2 = \frac{1}{8} m A^2 e^{-\beta t} \left[\beta^2 \cos^2(\omega_1 t - \delta) + 4\omega_1^2 \sin^2(\omega_1 t - \delta) \right]^2$$

$$E = T + U$$

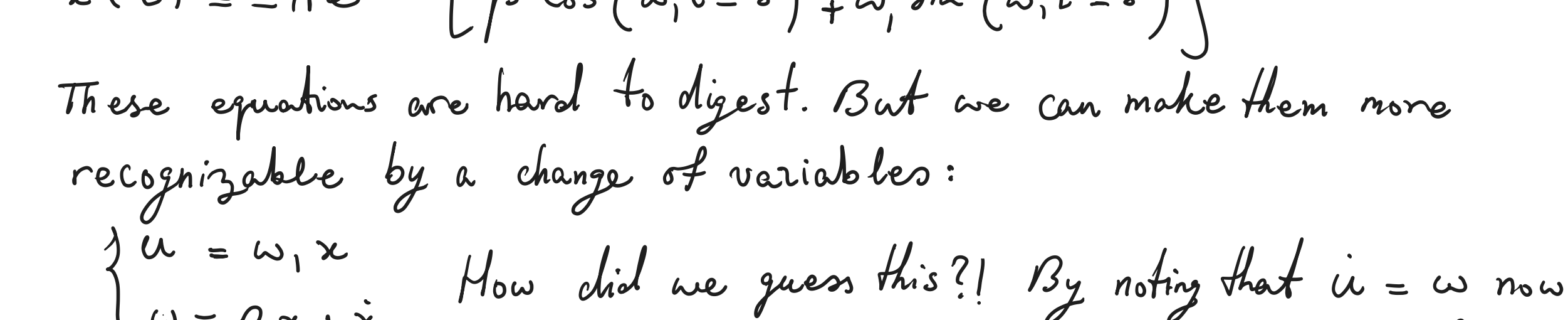
$$= \frac{1}{8} m A^2 e^{-\beta t} \left(\left[\beta \cos(\omega_1 t - \delta) + 2\omega_1 \sin(\omega_1 t - \delta) \right]^2 + 4\omega_0^2 \cos^2(\omega_1 t - \delta) \right)$$

$$= \frac{1}{8} m A^2 e^{-\beta t} \left(4\omega_0^2 + 2\beta\omega_1 \sin(2\omega_1 t - 2\delta) + \beta^2 \cos^2(2\omega_1 t - 2\delta) \right)$$

$$= \frac{1}{2} m A^2 \omega_0^2 e^{-\beta t} \left[1 + \frac{\beta}{2\omega_0} \cos(2\omega_1 t - 2\delta') \right] \quad \text{with } \delta' = \tan^{-1}\left(\frac{2\omega_1}{\beta}\right)$$

It is obvious that E is modulated by the exponential $e^{-\beta t}$, which is decaying over time.

Note that when $\beta \rightarrow 0$ (no damping) $E \rightarrow \frac{1}{2} m A^2 \omega_0^2$ which is the energy of a simple harmonic oscillator.



Problem 7) Recall again the eqns for an underdamped motion:

$$x(t) = A e^{-\beta t} \cos(\omega_1 t - \delta) \quad \text{where } \omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

$$\dot{x}(t) = -A e^{-\beta t} \left[\beta \cos(\omega_1 t - \delta) + \omega_1 \sin(\omega_1 t - \delta) \right]$$

These equations are hard to digest. But we can make them more recognizable by a change of variables:

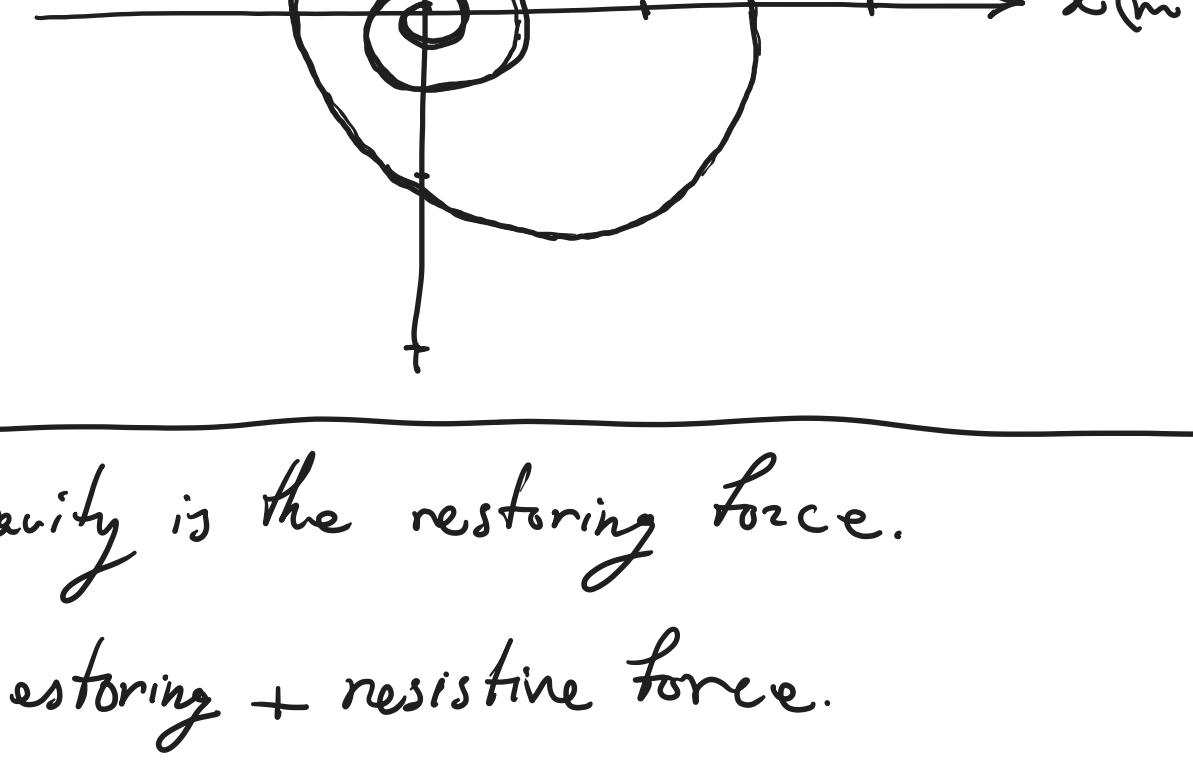
$$\begin{cases} u = \omega_1 x \\ \omega = \beta x + \dot{x} \end{cases} \quad \text{How did we guess this? By noting that } u = \omega \text{ now in a simple spiral form.}$$

$$\Rightarrow \begin{cases} u = \omega_1 A e^{-\beta t} \cos(\omega_1 t - \delta) \\ \omega = -\omega_1 A e^{-\beta t} \sin(\omega_1 t - \delta) \end{cases} \Rightarrow u^2 + \omega^2 = \omega_1^2 A^2 e^{-2\beta t}$$

To make the inward spiral motion eqn clearer, define $\phi = \omega_1 t$

$$\Rightarrow \phi = \omega_1 A e^{-\beta t} \cos(\phi - \delta)$$

Because $(x, \dot{x}) \rightarrow (u, \omega)$ transformation is linear, the phase diagram in $(x, \dot{x})$ plane will look similar to phase diagram in (u, ω) plane, perhaps only skewed or scaled. The actual shape is like the following:



Problem 8) Here gravity is the restoring force.

$$F = m(l\ddot{\theta}) = \text{Restoring} + \text{resistive force.}$$

$$m l \ddot{\theta} = -m g \sin \theta - 2m \sqrt{\frac{g}{l}} (l \dot{\theta}) \quad ; \quad \text{Assuming small oscillations, } \sin \theta \approx \theta$$

$$\Rightarrow \ddot{\theta} + 2\sqrt{\frac{g}{l}} \dot{\theta} + \frac{g}{l} \theta = 0 \Rightarrow \omega^2 = \frac{g}{l}; \quad \beta = \sqrt{\frac{g}{l}} = \omega_0$$

Thus, the pendulum is critically damped, which has the solution,

$$x(t) = (A + Bt) e^{-\beta t}; \quad \text{We use the initial condition,}$$

$$\theta(t=0) = \alpha \quad \text{and} \quad \dot{\theta}(t=0) = 0 \quad \text{to determine } A \text{ \& } B.$$

$$\theta(t) = B e^{-\beta t} - \beta(A + Bt) e^{-\beta t} \Rightarrow \begin{cases} \theta(t=0) = \alpha = A \\ \dot{\theta}(t=0) = 0 = B - \beta A \Rightarrow B = \alpha \beta \end{cases}$$

$$\Rightarrow \begin{cases} \theta(t) = \alpha \left(1 + \sqrt{\frac{g}{l}} t \right) e^{-\sqrt{\frac{g}{l}} t} \\ \dot{\theta}(t) = \frac{-\alpha g}{l} t e^{-\sqrt{\frac{g}{l}} t} \end{cases}$$

Problem 9) The complementary solution $x_c(t)$ is the same as before for the damped oscillation.

$$x(t) = e^{-\beta t} \left[A_1 \exp(\sqrt{\beta^2 - \omega_0^2} t) + A_2 \exp(-\sqrt{\beta^2 - \omega_0^2} t) \right]$$

For the particular solution, we can try,

$$x_p(t) = D \cos(\omega t - \delta)$$

Substituting $x_p(t)$ in the motion eqn; $\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = A \cos \omega t$

$$\text{leads to, } \begin{cases} A - D[(\omega_0^2 - \omega^2) \cos \delta + 2\omega \beta \sin \delta] \cos \omega t \\ -D[(\omega_0^2 - \omega^2) \sin \delta - 2\omega \beta \cos \delta] \sin \omega t = 0 \end{cases}$$

Because $\sin$ and $\cos$ are linearly independent functions, the coefficient of each must be zero. For the $\sin \omega t$ term:

$$\tan \delta = \frac{2\omega \beta}{\omega_0^2 - \omega^2}$$

$$\Rightarrow \begin{cases} \sin \delta = \frac{2\omega \beta}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}} \\ \cos \delta = \frac{\omega_0^2 - \omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}} \end{cases}$$

For the $\cos \omega t$ term, we have

$$D = \frac{A}{(\omega_0^2 - \omega^2) \cos \delta + 2\omega \beta \sin \delta}$$

$$= \frac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}}$$

$$\Rightarrow x_p(t) = \frac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}} \cos(\omega t - \delta) \quad \text{with } \delta = \tan^{-1}\left(\frac{2\omega \beta}{\omega_0^2 - \omega^2}\right)$$

where δ represents the phase difference between the driving force and the resultant motion. The general solution is

$$x(t) = x_c(t) + x_p(t)$$

where $x_c(t)$ represents the transient effects (that die out because of its $e^{-\beta t}$ factor) and $x_p(t)$ represents the steady-state effects,

$$x(t \gg 1/\beta) = x_p(t)$$

The transient effect although not permanent is very important in electrical circuits.

Problem 10) We would like to know the kinetic energy resonance, that is, the frequency ω at which T is maximum.

$$T = \frac{1}{2} m \dot{x}^2; \quad x_p(t) = \frac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}} \cos(\omega t - \delta)$$

Although the complete solution is the sum of x_c and x_p , we can safely ignore $x_c(t)$ because it is only a transitive effect that will vanish over time.

$$\Rightarrow \dot{x} = \dot{x}_p(t) = \frac{-A \omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}} \sin(\omega t - \delta)$$

$$\Rightarrow T = \frac{m A^2}{2} \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2} \sin^2(\omega t - \delta)$$

This T varies over time. Thus, we need to compute the average T over one period to find ω_E .

$$\Rightarrow \langle T \rangle = \frac{m A^2}{2} \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2} \langle \sin^2(\omega t - \delta) \rangle$$

$$\langle \sin^2(\omega t - \delta) \rangle = \frac{\omega}{2\pi} \int_0^{2\pi} \sin^2(\omega t - \delta) dt = \frac{1}{2}$$

$$\Rightarrow \langle T \rangle = \frac{m A^2}{4} \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\omega^2 \beta^2}$$

$$\frac{d\langle T \rangle}{d\omega} = 0 \Rightarrow \boxed{\omega_E = \omega_0} \quad \text{which is different from}$$

the amplitude resonance frequency $\omega_A = \sqrt{\omega_0^2 - 2\beta^2}$.

Because the potential energy directly depends on the amplitude, the potential energy resonance frequency must also occur at ω_A .

The fact that resonances occur at different frequencies is not surprising because the system is constantly dissipating energy into the medium.

Problem 11) Mechanical Force (mg) $\equiv$ Electromagnetic force (emf)

The damping parameter $b \equiv$ electrical resistance

The displacement $x \equiv$ electrical charge q .

$$1/k \equiv C, \quad m \rightarrow L, \quad F \rightarrow \mathcal{E}, \quad \dot{x} \rightarrow I$$

The equilibrium point x for the mass m

$$\text{can be obtained as, } \Sigma F = 0 = mg - kx \Rightarrow h = \frac{mg}{k}$$

Thus, the equation of motion is $m\ddot{x} + k(x-h) = 0$ or $m\ddot{x} - kx = kh$

whose solution is $x(t) = h + A \cos \omega_0 t$ with $\{x(t=0) = h + A$

The equivalent electrical circuit becomes, $\{i(t=0) = 0$

$$\textcircled{1} \quad L \frac{dI}{dt} + \frac{1}{C} \int I dt = \mathcal{E} = \frac{\mathcal{E}_0}{C} \quad \text{with } q_C \text{ representing the}$$

charge that must be applied to the capacitor to produce a voltage $\mathcal{E}$. Since $I \equiv \dot{q} \Rightarrow L \dot{q} + \frac{q}{C} = \frac{\mathcal{E}_0}{C}$

For $q(t=0) = q_0$ and $\dot{q}(t=0) = 0$ the solution becomes:

$$q(t) = q_C + (q_0 - q_C) \cos \omega_0 t$$

Note that we used Kirchhoff's loop rule to derive eqn $\textcircled{1}$:

$$\Sigma V_i = 0. \quad \text{For the solenoid } V_L(t) = L \frac{dI}{dt} = L \dot{q}$$

$$\text{For the capacitor, } C = \frac{dQ}{dV} \Rightarrow dV_C(t) = \frac{dQ}{C} = \frac{q(t)}{C}$$

$$V_L + V_C = \mathcal{E} = \frac{\mathcal{E}_0}{C}$$

Problem 12) $V(x) = V(x_0) + V'(x_0)(x-x_0) + \frac{1}{2} V''(x_0)(x-x_0)^2 + \frac{1}{3!} V'''(x_0)(x-x_0)^3 + \dots$

For equilibrium, $V(x_0) = 0 \Rightarrow V(x) \approx \frac{1}{2} V''(x_0)(x-x_0)^2$

This is the Hooke's law! $F = -\frac{dV}{dx}$ and we know $\omega = \sqrt{\frac{V''(x_0)}{m}}$

With this in mind, Let's check the potential $V(x) = \frac{A}{x^2} - \frac{B}{x}$

with $A, B > 0 \Rightarrow V'(x) = -\frac{2A}{x^3} + \frac{B}{x^2}$; An equilibrium point requires $V'(x) = 0 \Rightarrow \boxed{x_0 = \frac{2A}{B}}$ the equilibrium point.

$$V''(x) = \frac{6A}{x^4} - \frac{2B}{x^3} \Big|_{x=x_0} = \frac{2A}{B} \Rightarrow \omega = \sqrt{\frac{V''(x_0)}{m}} = \sqrt{\frac{B^3}{8mA^2}}$$

Problem 13) $L = T - U$; The kinetic energy is composed of radial and tangential components,

$$T = \frac{1}{2} m (\dot{x}^2 + (l + x)^2 \dot{\theta}^2) \quad \textcircled{1}$$

The potential energy has gravity and tangential speed

$$\text{spring components; } U(x, \theta) = -mg(l+x) \cos \theta + \frac{1}{2} k x^2 \quad \textcircled{2}$$

$$\textcircled{1} \textcircled{2} \Rightarrow L = \frac{1}{2} m (\dot{x}^2 + (l+x)^2 \dot{\theta}^2) + mg(l+x) \cos \theta - \frac{1}{2} k x^2$$

There are two variables here: (x, θ)

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow m \ddot{x} = m(l+x) \dot{\theta}^2 + mg \cos \theta - kx \quad \textcircled{3} \\ \frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = 0 \Rightarrow \frac{d}{dt} (m(l+x)^2 \dot{\theta}) = -mg(l+x) \sin \theta \quad \textcircled{4} \end{cases}$$

$$\Rightarrow m(l+x) \ddot{\theta} + 2m(l+x) \dot{x} \dot{\theta} = -mg(l+x) \sin \theta$$

$$\Rightarrow m(l+x) \ddot{\theta} + 2m \dot{x} \dot{\theta} = -mg \sin \theta \quad \textcircled{5}$$

It is simply the Newton's law $F = ma$ with the second term being related to the centripetal acceleration $-(l+x) \dot{\theta}^2$.

$\textcircled{4}$ states that torque equals the rate of change of angular momentum.

Problem 14) $x(t) = A \cos(\omega t)$ $\textcircled{1}$

$$\Rightarrow (x, y)_m = (x + l \sin \theta, -l \cos \theta)$$

$$\Rightarrow V_m^2 = \dot{x}^2 + \dot{y}^2 = l^2 \dot{\theta}^2 + \dot{x}^2 + 2l \dot{x} \dot{\theta} \cos \theta$$

$$\Rightarrow L = \frac{1}{2} m (l^2 \dot{\theta}^2 + \dot{x}^2 + 2l \dot{x} \dot{\theta} \cos \theta) + mgl \cos \theta$$

$$\frac{\partial L}{\partial \theta} = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} \Rightarrow \frac{d}{dt} (ml^2 \dot{\theta} + ml \dot{x} \cos \theta) = -ml \dot{x} \sin \theta - mgl \sin \theta$$

$$\Rightarrow l \ddot{\theta} + \dot{x} \cos \theta = -g \sin \theta \quad \textcircled{2}$$

$$\dot{x}, \ddot{x} \Rightarrow l \ddot{\theta} - A \omega^2 \cos(\omega t) \cos \theta + g \sin \theta = 0$$

Assuming $\sin \theta \approx \theta \Rightarrow \ddot{\theta} + \omega_0^2 \theta = a \omega^2 \cos(\omega t)$; $\omega_0 = \sqrt{\frac{g}{l}}$; $a = \frac{A}{l}$

which implies a driven oscillation where we have also used,

$$\cos \theta \Big|_{\theta \rightarrow 0} = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots$$

$$\Rightarrow \theta(t) = \frac{a \omega^2}{\omega_0^2 - \omega^2} \cos(\omega t) + B \cos(\omega_0 t - \phi) \quad \text{with } B \text{ and } \phi \text{ given by initial conditions. What happens if } \omega \rightarrow \omega_0?$$

